

ECMA 30760 Summary

2 Voting

2.1 Introduction

Lemma 2.1.1. *If X and Y are dominating sets, then $X \subseteq Y$ or $Y \subseteq X$.*

2.2 Gibbard–Satterthwaite

Theorem 2.2.1 (Gibbard–Satterthwaite). *If $|A| > 2$, then every surjective and strategy-proof social choice function is dictatorial, in that there is some agent i such that*

$$f(\succ) = \arg \max_{\succ_i} A$$

for all $\succ$. (Here, the $\arg \max$ over $\succ_i$ means that the ordering of the right hand side is determined by $\succ_i$.)

3 Aggregation

3.1 Ordinal Aggregation

Theorem 3.1.1 (Arrow’s Impossibility). *If $|A| > 2$, then the only (weak or strict) social welfare functions that satisfy IIA and the Paretian property are dictatorial.*

3.2 Cardinal Aggregation

Theorem 3.2.1 (Debreu). *For any $\succsim_s$ satisfying continuity and independence of unconcerned agents, there exist functions $\{g_i\}_{i \in N}$ such that*

$$u \succsim_s u' \iff \sum_i g_i(u_i) \geq \sum_i g_i(u'_i).$$

Theorem 3.2.2. *If $\succsim_s$ satisfies continuity, independence of unconcerned agents, strict monotonicity, and scale invariance, and if $|N| \geq 2$, then there are positive weights $w_1, \dots, w_{|N|}$ which sum to 1 and some $\gamma \in \mathbb{R}$ such that*

$$u \succsim_s v \iff \sum_i w_i \frac{u_i^{1-\gamma} - 1}{1-\gamma} \geq \sum_i w_i \frac{v_i^{1-\gamma} - 1}{1-\gamma}.$$

3.3 Aggregation Under Uncertainty

Theorem 3.3.1 (Harsanyi). *The planner’s utility U_s is Paretian iff there is a constant β and weights $(w_i)_{i \in N}$ which are nonnegative such that*

$$u_s(\theta) = \beta + \sum_i w_i u_i(\theta), \quad \forall \theta \in \Theta.$$

Equivalently, we have a necessary and sufficient condition for the Paretian property of

$$U_s(p) = \beta + \sum_i w_i U_i(p).$$

4 Social Choice with Money

4.1 Introduction

Proposition 4.1.1. *The second-price transfer payment mechanism is strategy-proof.*

4.2 Quasilinear Preferences

Proposition 4.2.1 (Quasilinear representation of preferences). *If $\succsim$ satisfies the axioms, there exists a utility function $u : A \rightarrow \mathbb{R}$ such that*

$$(a, p) \succsim (a', p') \iff u(a) - p \geq u(a') - p'.$$

4.3 Implementability, Cyclical Monotonicity, and Farkas' Lemma

Proposition 4.3.1 (Equivalence of cyclical monotonicity and implementability). *A social choice function $g : V^N \rightarrow A$ is implementable in dominant strategies iff g satisfies cyclical monotonicity.*

Proposition 4.3.2. *Suppose $g : \mathcal{U}^{|N|} \rightarrow A$ is such that*

$$g(u_1, u_2, \dots, u_{|N|}) \in \arg \max_{a \in A} \sum_{i \in N} u_i(a).$$

Then g is cyclically monotone.

Corollary 4.3.3. *Pairing this result with the referenced proposition yields that any welfare-maximizing social choice function is implementable in dominant strategies.*

4.5 The Vickrey–Clarke–Groves Mechanism

Proposition 4.5.1. *Let g be a welfare-maximizing utility function. Consider the rule*

$$p_i(u_i, u_{-i}) := - \sum_{j \neq i} u_j(g(u_i, u_{-i})).$$

Under these transfer payments, truthful reporting is a dominant strategy; p implements g .

Proposition 4.5.2. *If $h : \mathcal{U}^{|N|-1} \rightarrow \mathbb{R}$ and p implements g , then if*

$$p'_i(u_i, u_{-i}) = p_i(u_i, u_{-i}) + h(u_{-i}),$$

$p' = (p'_i, p_{-i})$ also implements g .

Corollary 4.5.3. *In particular, the payments in the VCG mechanism implement the VCG mechanism's social choice function.*

Theorem 4.5.4 (Roberts'). *Suppose that $|A| \geq 3$. $g : \mathcal{U}^{|N|} \rightarrow A$ is implementable in dominant strategies iff there are nonnegative weights $\{w_i\}_{i \in N}$ which are not identically zero and constants $\{C_a\}_{a \in A}$ for each $a \in A$ such that*

$$g(u_1, \dots, u_{|N|}) \in \arg \max_{a \in A} \sum_{i \in N} w_i u_i(a) + C_a.$$

4.6 Domain Restrictions

Theorem 4.6.1 (Saks and Yu, 2005). *Suppose $\mathcal{V} \subseteq \mathcal{U}$ is convex. $g : \mathcal{V}^{|N|} \rightarrow A$ is implementable in dominant strategies iff it is weakly monotonic.*

5 Social Choice with Indifferences, Constrained Allocation

5.1 Introduction and the Equivalence of Constrained Allocation

Proposition 5.1.1 (Equivalence of social choice with indifference and constrained allocation). *Every constrained allocation problem can be written as a social choice problem with indifference. Conversely, every social choice problem with indifference can be written as a constrained allocation problem. Under these translations, mechanisms are the same up to outcomes that all agents regard as indifferent.*

5.2 Two Agent Constrained Allocation

Proposition 5.2.1 (Two-agent local dictatorships). *Let G be the graph whose nodes are infeasible allocations and whose edges connect two infeasible allocations that share a coordinate. After removing objects that can never be assigned to an agent in any feasible allocation, a mechanism f is strategy-proof and Pareto efficient if and only if it is a local dictatorship: each connected component of G is assigned a local dictator, and for every profile with top-ranked objects (x, y) , the mechanism gives both agents their top choices if $(x, y) \in C$; otherwise, the local dictator for the component containing (x, y) keeps that agent's top choice and the other agent receives her favorite object compatible with the dictator's top choice.*

Lemma 5.2.2. *Suppose that there exists $x \in X_i$ for some i such that $\mu(i) \neq x$ for every $\mu \in C$. If f is strategy-proof and Pareto efficient, then for $\succ_i$ and $\succ'_i$ that differ only in the position of x , then*

$$f(\succ_i, \succ_j)(i) = f(\succ'_i, \succ_j)(i).$$

Furthermore, j 's object is unchanged by i 's misreport:

$$f(\succ_i, \succ_j)(j) = f(\succ'_i, \succ_j)(j).$$

Lemma 5.2.3. *For any strategy-proof and efficient mechanism f , each infeasible allocation $(x, y) \notin C$ is associated with a local dictator. In particular, there exists a function $D : \overline{C} \rightarrow \{1, 2\}$ that describes who is the local dictator at every infeasible allocation. Furthermore, if $(x, y'') \notin C$ as well, then*

$$D((x, y)) = 1 \implies D((x, y'')) = 1.$$

6 Auctions

6.1 Optimal Posted Price With a Single Buyer

Proposition 6.1.1. *In this single-buyer, single-good environment with a posted price, the optimal price is the p^* satisfying*

$$p^* = \frac{1 - F(p^*)}{f(p^*)}.$$

- The optimal price leads to an inefficient outcome, and
- The optimal price depends on F .

6.2 Overview of Auctions

Proposition 6.2.1 (Revenue of a second price auction). *The revenue of a second-price auction is*

$$R_{SPA} = 1 - \int_0^1 F_{(2)}(w)dw = \int_0^1 (1 - F(w))^2 dw.$$

Proposition 6.2.2 (Revenue of a first-price auction). *An equilibrium bidding strategy in a symmetric first price auction with independent value distribution with CDF F for a bidder with value v is*

$$b(v) = v - \frac{1}{F(v)} \int_0^v F(w)dw.$$

The expected revenue is

$$R_{FPA} = 1 - \int_0^1 F_{(2)}(w)dw = \int_0^1 (1 - F(w))^2 dw,$$

which is exactly the expected revenue of the second-price auction.

6.3 Revelation Principle

Proposition 6.3.1 (Revelation Principle). *For any G and equilibrium s , there exists a direct mechanism in which it is a Bayes–Nash equilibrium for agents to truthfully report such that the outcome (set of allocation probabilities and transfer payments) is the same under G, s and under the direct mechanism.*

6.4 Revenue-Maximizing Auctions

Proposition 6.4.1. *If s is a Bayes–Nash equilibrium of an auction game G , then the induced interim allocation probabilities X_i and payments P_i satisfy the following:*

- (i) *The interim allocation probability $X(\cdot)$ is nondecreasing in the report.*
- (ii) *For any v ,*

$$P(v) = P(0) + vX(v) - \int_0^v X(t)dt.$$

The only choices that the seller has are in controlling $P(0)$ and $X(v)$ for each $v \in [0, 1]$. The payment at any value is then uniquely determined.

Corollary 6.4.2 (A revenue-maximizing auction's payments). *A revenue-maximizing auction satisfying individual rationality has payment schedule*

$$P(v) = vX(v) - \int_0^v X(t)dt.$$

This payment schedule and $X_i(\cdot)$ being nondecreasing (from the referenced proposition) are necessary conditions for an auction to be revenue-maximizing. Note that we have not yet determined what exactly $X(\cdot)$ is in the revenue-maximizing auction.

Proposition 6.4.3. *For any G, G' and equilibria s, s' such that X_G and $X_{G'}$ are the same and $P_G(0) = P_{G'}(0)$, the revenue is the same.*

Theorem 6.4.4 (Revenue maximization of the virtual value auction). *Consider individually rational auctions with a single good and symmetric bidders whose values are drawn iid from a regular distribution F . One¹ revenue-maximizing auction has assignment probabilities for each bidder i given by*

$$x_i(v_i, v_{-i}) = \begin{cases} 0, & c(v_i) \leq 0 \text{ or } c(v_i) < c(v_j) \text{ for some } j \\ \frac{1}{m}, & m := |\{j : v_j = v_i\}| \end{cases}.$$

Moreover, the direct mechanism with this allocation rule and payments

$$p_i(v_i, v_{-i}) = v_i x_i(v_i, v_{-i}) - \int_0^{v_i} x_i(t, v_{-i}) dt$$

is a revenue-maximizing virtual value auction.

6.5 Multiple Goods and a Single Buyer

Proposition 6.5.1. *In a two-good, single buyer auction with independent value distributions $F^1, F^2 \sim \text{Unif}(0, 1)$, the optimal mechanism is one in which the buyer has the options to:*

- (i) *Buy either good at price $\frac{3}{4}$, or*
- (ii) *Buy both goods at price $\frac{4-\sqrt{2}}{3}$.*

Proposition 6.5.2. *If both F^1 and F^2 are Beta distributions, the optimal mechanism offers an uncountably large menu.*

7 Monopolistic Screening

7.1 Mussa–Rosen (1978)

¹Any revenue-maximizing auction must have $x_i(v_i, v_{-i}) = 0$ for any i where $c(v_i) \leq 0$ or $c(v_i) < c(v_j)$ for some j .

Proposition 7.1.1. Assume F is regular, so its virtual value

$$c(\theta) := \theta - \frac{1 - F(\theta)}{f(\theta)}$$

is strictly increasing. Then an individually rational revenue-maximizing direct mechanism has quality rule

$$q^*(\theta) = \begin{cases} 0, & c(\theta) \leq C'(0) \\ (C')^{-1}(c(\theta)), & C'(0) < c(\theta) < C'(\bar{q}) \\ \bar{q}, & c(\theta) \geq C'(\bar{q}) \end{cases}$$

and payment rule

$$p^*(\theta) = \theta q^*(\theta) - \int_0^\theta q^*(t) dt.$$

In particular, q^* is nondecreasing.

8 Cooperative Games

8.2 Cooperative Games with Transferable Utility

Theorem 8.2.1 (Bonderava–Shapley). A cooperative game with transferable utility (N, v) has a nonempty core if and only if for every set of balanced weights δ ,

$$\sum_{S \subseteq N, |S| \neq \emptyset} \delta(S) v(S) \leq v(N).$$

9 Stable Matching

9.1 One-Sided Matching

Proposition 9.1.1. A matching μ is in the core if and only if it is stable. In particular, to check the core, we just need to check deviations from coalitions of size 2.

9.2 One-to-One Two-Sided Matching

Proposition 9.2.1. The M -proposing DA outcome is stable. Hence, the core of the one-to-one two-sided matching problem is nonempty.

Proposition 9.2.2. In the M -proposing DA algorithm, each $m \in M$ who is matched is matched with his favorite achievable partner and each $w \in W$ who is matched is matched with her least-favorite achievable partner.

Proposition 9.2.3. There are no stable and strategy-proof one-to-one two-sided matching mechanisms.

Theorem 9.2.4. In the mechanism that returns the M -proposing DA outcome for every preference profile, it is a dominant strategy for each $m \in M$ to truthfully report his preference.

Lemma 9.2.5 (Blocking). Let μ_{DA} be the outcome of the M -proposing DA mechanism for the preferences $\succ$. Let μ be any other matching (not necessarily stable). Let M' be the set of men who strictly prefer their matches under μ to their matches under μ_{DA} , i.e.

$$M' = \{m \in M : \mu(m) \succ_m \mu_{DA}(m)\}.$$

Then

(i) $M' \neq M$, and

(ii) If $M' \neq \emptyset$, then there is an $m \in M \setminus M'$ and a w in $\mu(M')$ such that (m, w) is a blocking pair for μ .

9.3 Many-to-One Two-Sided Matching

Proposition 9.3.1 (Stability of DA-M under responsiveness.). *The many-to-one two-sided DA-M algorithm is stable if each w 's preferences are responsive.*

10 Allocation Problems

10.1 Deterministic Allocation

Svensson (1999)

Lemma 10.1.1. *For any mechanism f , the following are equivalent:*

- (i) f is group strategy-proof.
- (ii) f is strategy-proof and non-bossy.
- (iii) f is monotonic.

Lemma 10.1.2. *If a mechanism is group strategy-proof, then it is Pareto efficient iff it is surjective.*

Theorem 10.1.3 (Svensson). *A mechanism f is group strategy-proof and neutral iff it is a serial dictatorship.*

Top Trading Cycles

Proposition 10.1.4. *The top trading cycles mechanism is group strategy-proof and Pareto efficient.*

10.2 Random Allocation

Theorem 10.2.1 (Birkhoff Von–Neumann). *Any bistochastic matrix can be written as a convex combination of permutation matrices.*

Proposition 10.2.2. $RSD = RTTC$.

Proposition 10.2.3. *RSD and RTTC are strategy-proof, ex-post efficient, and satisfy ETOE. Furthermore, by Bastek and Ehlers (2024), these three properties uniquely characterize RSD/RTTC up until $|N| = 4$, while there exist strategy-proof, ex-post efficient, and ETOE mechanisms for $|N| \geq 5$.*

Proposition 10.2.4. *The probabilistic serial mechanism is ordinally efficient.*

Theorem 10.2.5 (Bogomolnaia and Moulin (2001)). *There is no strategy-proof, ordinally efficient mechanism that satisfies equal treatment of equals.*

11 Information Design

11.2 General Model

Lemma 11.2.1. *A necessary condition for a distribution over posteriors to have some signal structure that induces it is that the expectation of the posterior beliefs is the prior.*

Proposition 11.2.2. *There is a signal structure with value V^* iff there is a Bayes-plausible distribution over posterior beliefs such that*

$$\int \hat{u}_S(\mu) d\tau(\mu) = V^*.$$

Theorem 11.2.3 (Carathéodory). *Suppose $X \subseteq \mathbb{R}^d$ and y is a point in the convex hull of X . Then there are $d + 1$ points in X such that y is in the convex hull of those points.*

A Tools

A.3 Farkas' Lemma

Lemma A.3.1 (Farkas'). Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Either

(i) There exists $x \in \mathbb{R}^n$ such that

$$Ax \leq b.$$

(ii) There exists² $y \in \mathbb{R}^m$ with $y \geq 0$ such that

$$yA = 0, \quad y \cdot b < 0.$$

but not both.

Lemma A.3.2 (Rational Farkas'). Let $A \in \mathbb{Q}^{m \times n}$ and $b \in \mathbb{Q}^m$. Either

(i) There exists $x \in \mathbb{Q}^n$ such that

$$Ax \leq b.$$

(ii) There exists $y \in \mathbb{Z}^m$ with $y \geq 0$ such that

$$yA = 0, \quad y \cdot b < 0.$$

but not both.

Lemma A.3.3 (Farkas' II). Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $B \in \mathbb{R}^{p \times n}$, and $c \in \mathbb{R}^p$. Then either

(i) There exists $x \in \mathbb{R}^n$ such that

$$Ax \leq b, \quad Bx = c.$$

(ii) There exists $s \in \mathbb{R}^m$ and $t \in \mathbb{R}^p$ such that

$$s \geq 0, \quad sA + tB = 0, \quad s \cdot b + t \cdot c < 0.$$

but not both.

Lemma A.3.4 (Farkas' III). Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $B \in \mathbb{R}^{p \times n}$, and $c \in \mathbb{R}^p$. Then either

(i) There exists $x \in \mathbb{R}^n$ with $x \geq 0$ such that

$$Ax \leq b, \quad Bx = c.$$

(ii) There exists $s \in \mathbb{R}^m$ and $t \in \mathbb{R}^p$ with $s \geq 0$ such that

$$sA + tB \geq 0, \quad s \cdot b + t \cdot c < 0.$$

but not both.

Lemma A.3.5 (Gordan's). Let $A \in \mathbb{Q}^{m \times d}$. Then either

(i) There exists $y \in \mathbb{R}^d$ such that

$$Ay \gg 0,$$

i.e. every entry of Ay is strictly positive, or

(ii) There exists a nonzero vector of nonnegative integers $z \neq 0$ such that

$$zA = 0.$$

but not both.

A.4 Minimax Theorem

Theorem A.4.1 (Minimax). In a two-player zero-sum game,

$$\max_{\alpha_1 \in \Delta A_1} \min_{\alpha_2 \in \Delta A_2} \alpha_1 B \alpha_2 = \min_{\alpha_2 \in \Delta A_2} \max_{\alpha_1 \in \Delta A_1} \alpha_1 B \alpha_2.$$

²Here, and in the rest of the chapter, we take $y \in \mathbb{R}^m$ to be a row vector. We call y the *multiplier vector* or *certificate of infeasibility*.

A.5 The Standard Form

Proposition A.5.1. *The system $Ax \leq b$ has a solution if and only if*

$$(A \quad -A \quad I) \begin{pmatrix} y \\ y' \\ s \end{pmatrix} = b$$

has a solution with $y, y', s \geq 0$.

A.6 Polyhedra

Proposition A.6.1. *Let $v \in P(A, b)$. The following are equivalent:*

- (i) *v is a vertex of $P(A, b)$.*
- (ii) *There do not exist distinct $x, y \in P(A, b)$ and $\theta \in (0, 1)$ such that*

$$v = \theta x + (1 - \theta)y.$$

- (iii) *There is no nonzero vector d such that both $v + d$ and $v - d$ belong to $P(A, b)$.*
- (iv) *There are n linearly independent rows $A_{i_1}, \dots, A_{i_n}$ of A such that*

$$A_{i_1}v = b_{i_1}, \quad \dots, \quad A_{i_n}v = b_{i_n}.$$

Proposition A.6.2. *Every polytope is the convex hull of its vertices. Conversely, the convex hull of any finite subset of $\mathbb{R}^n$ is a polytope.*

Theorem A.6.3 (Birkhoff–von Neumann). *The vertices of B_n are exactly the permutation matrices.*

A.7 Lattices and Tarski's Fixed Point Theorem

Theorem A.7.1 (Tarski). *If $(L, \geq)$ is a complete lattice and $f : L \rightarrow L$ is order-preserving, then the set of fixed points*

$$F = \{x \in L : f(x) = x\}$$

is a nonempty complete lattice under $\geq$.

Definitions

2 Voting

2.1 Introduction

Definition 2.1.1 (Social choice function). A social choice function is a mapping $f : P^N \rightarrow A$.

Definition 2.1.2 (Common Voting Rules). • Plurality: Choose whichever outcome is top-ranked by the most people. (Along with some tiebreaking rule.)

- Instant runoff: All outcomes but the two top-ranked outcomes³ are dropped, then the winner is decided by plurality.
- Elimination: The outcome that is top-ranked by the fewest people is removed and the process repeats.
- Borda count: Let $n = |A|$. Give n points to a candidate for each agent that top-ranks them, $n - 1$ points for each agent that second-ranks them, and so on. The outcome with the most points wins.
- Summation social choice functions: Assign points $\phi_i(m)$ for agent i 's m th ranked outcome, and the outcome with the most points wins.
- Pairwise comparison graph: We arrange the outcomes into a directed graph with all possible edges labeled by the number of people who prefer the tail node to the head node. A special case is the Copeland social choice function, which counts for each outcome how many arrows point out, and the winner is the node with the highest count. See the referenced figure.

Definition 2.1.3 (Condorcet winner, Condorcet property). An outcome x is a Condorcet winner if it beats all other outcomes in a head-to-head election. A social choice function satisfies the Condorcet property if it picks a Condorcet winner, when one exists.

Definition 2.1.4 (Dominating set of outcomes, Smith set, Smith property). A set of outcomes $A' \subseteq A$ is dominating if it is nonempty and for all $x \in A', y \in A \setminus A', x$ beats y in a head-to-head election.

The Smith set is the smallest dominating set. A social choice function f satisfies the Smith property if it always chooses something in the Smith set.

2.2 Gibbard–Satterthwaite

Definition 2.2.1 (Strategy-proof). A social choice function is strategy-proof if there is no preference profile $(\succ_1, \dots, \succ_{|N|})$, agent i , and preference $\succ'_i$ such that

$$f(\succ_1, \dots, \succ_{i-1}, \succ'_i, \succ_{i+1}, \dots, \succ_{|N|}) \succ_i f(\succ_1, \dots, \succ_{i-1}, \succ_i, \succ_{i+1}, \dots, \succ_{|N|}).$$

Definition 2.2.2 (Lower contour set). Given a preference $\succ_i$ on A and $a \in A$, define the lower contour set as $L_{\succ_i}(a) = \{b : a \succ_i b\}$, which is the set of outcomes dispreferred to a .

Definition 2.2.3 (Monotonic social choice function). f is monotonic if for any preference profiles $\succ, \succ'$ such that for every i , $L_{\succ'_i}(f(\succ)) \supseteq L_{\succ_i}(f(\succ))$, we have $f(\succ') = f(\succ)$.

Definition 2.2.4 (Pareto efficient social choice function). f is Pareto efficient if for all preference profiles $\succ$, $a \succ_i b$ for all i implies that $f(\succ) \neq b$.

3 Aggregation

3.1 Ordinal Aggregation

Definition 3.1.1 (Strict, weak social welfare function). A strict social welfare function is a function $F : P^N \rightarrow P$. A weak social welfare function is a function $F : P^N \rightarrow R$.

³The two outcomes that have the highest number of top-rank votes.

Definition 3.1.2 (Kendall–Tau distance). For any two strict rankings $\succ_i, \succ'_i$ on A , define the distance $d(\succ_i, \succ'_i)$ between the two rankings as the number of pairs $a, b \in A$ such that $a \prec_i b$ and $a \succ'_i b$ or $a \succ_i b$ and $a \prec'_i b$.

Definition 3.1.3 (Paretian property). A social welfare function F satisfies the Paretian property if for any $\succ$ and $a, b \in A$ is such that if for all i , $a \succ_i b$, then

$$aF(\succ)b.$$

Definition 3.1.4 (Independence of irrelevant alternatives). A social welfare function F satisfies independence of irrelevant alternatives (IIA) if for any two profiles $\succ, \succ'$ such that for all i , $\succ_i$ and $\succ'_i$ agree when restricted to $\{a, b\}$ for $a, b \in A$, then

$$aF(\succ)b \iff aF(\succ')b.$$

3.2 Cardinal Aggregation

Definition 3.2.1 (Cardinal social welfare ranking). A cardinal social welfare ranking is a complete, transitive and reflexive binary relation $\succsim_s$ on $\mathbb{R}_+^{|N|}$.

Definition 3.2.2 (Continuous cardinal social welfare ranking). We say that $\succsim_s$ is continuous if for any $u \in \mathbb{R}_+^{|N|}$, the sets

$$\{u' : u' \succsim_s u\}, \quad \{u' : u \succsim_s u'\}$$

are closed. In other words, if $\{u^n\} \rightarrow u^*$ is such that $u^n \succsim_s u'$ for each $n \in \mathbb{N}$, then $u^* \succsim_s u'$. Similarly, if $u' \succsim_s u^n$ for each $n \in \mathbb{N}$, then $u' \succsim_s u^*$.

Definition 3.2.3 (Independence of unconcerned agents). $\succsim_s$ is independent of unconcerned agents if for any $x, y \in \mathbb{R}_+$ and any agent i , we have

$$(x, u_{-i}) \succsim_s (x, u'_{-i}) \iff (y, u_{-i}) \succsim_s (y, u'_{-i})$$

for any u_{-i}, u'_{-i} .

Definition 3.2.4 (Strict monotonicity). We say that $\succsim_s$ is strictly monotonic if whenever $u_i \geq u'_i$ for all i and there is some j with $u_j > u'_j$, then $u \succ_s u'$.

Definition 3.2.5 (Scale invariance). We say $\succsim_s$ is scale invariant if

$$u \succsim_s u' \iff \lambda u \succsim_s \lambda u', \quad \forall \lambda \in \mathbb{R}_{>0}.$$

3.3 Aggregation Under Uncertainty

Definition 3.3.1 (Paretian property). The planner's utility function $U_s(\cdot)$ satisfies the Paretian property if $U_i(p) \geq U_i(q)$ for all i implies $U_s(p) \geq U_s(q)$.

4 Social Choice with Money

4.2 Quasilinear Preferences

Definition 4.2.1 (Quasilinear Preferences). A quasilinear preference v on $A \times \mathbb{R}$ can be written

$$v(a, p) = u(a) - p.$$

Definition 4.2.2 (Represents). Suppose $\succsim$ is a weak preference on $A \times \mathbb{R}$; i.e., it is complete, transitive, and reflexive. Suppose A is a second-countable⁴ topological space.

We say that $v : A \times \mathbb{R} \rightarrow \mathbb{R}$ represents $\succsim$ if

$$(a, p) \succsim (a', p') \iff v(a, p) \geq v(a', p').$$

4.3 Implementability, Cyclical Monotonicity, and Farkas' Lemma

⁴In other words, A 's topology is separable/has a countable base.

Definition 4.3.1 (Implementability). We say that a social choice function $g : \mathcal{U}^{|N|} \rightarrow A$ is implementable in dominant strategies if there exists $p : \mathcal{U}^{|N|} \rightarrow \mathbb{R}^n$ such that for every i, u_i, u_{-i} and u'_i ,

$$u_i(g(u_i, u_{-i})) - p_i(u_i, u_{-i}) \geq u_i(g(u'_i, u_{-i})) - p_i(u'_i, u_{-i}).$$

Equivalently,

$$p_i(u_i, u_{-i}) - p_i(u'_i, u_{-i}) \leq u_i(g(u_i, u_{-i})) - u_i(g(u'_i, u_{-i})).$$

Definition 4.3.2 (Cycles and cyclical monotonicity). Fix a player $i \in N$. A *cycle* in player i 's type space is a finite sequence

$$v^1, v^2, \dots, v^{q-1}, v^q = v^1,$$

where each $v^k \in V$. Given $v_{-i} \in V^{N \setminus \{i\}}$, this means that the reports of all players other than i are fixed at v_{-i} , while player i 's report moves around the cycle.

A social choice function $g : V^N \rightarrow A$ satisfies *cyclical monotonicity* if for every i , every v_{-i} , and every cycle in player i 's type space,

$$\sum_{k=1}^{q-1} v^k(g(v^k, v_{-i})) - v^k(g(v^{k+1}, v_{-i})) \geq 0.$$

4.5 The Vickrey–Clarke–Groves Mechanism

Definition 4.5.1 (Vickrey–Clarke–Groves (VCG) mechanism). The VCG mechanism has social choice function

$$g(u_1, \dots, u_{|N|}) \in \arg \max_{a \in A} \sum_i u_i(a)$$

and payments

$$p_i(u_i, u_{-i}) = \sum_{j \neq i} [u_j(a^*(u_{-i})) - u_j(g(u_i, u_{-i}))],$$

where $a^*(u_{-i}) \in \arg \max_{a \in A} \sum_{j \neq i} u_j(a)$.

4.6 Domain Restrictions

Definition 4.6.1 (Weak monotonicity). Let $\mathcal{V} \subseteq \mathcal{U}$. We say that $g : \mathcal{V}^{|N|} \rightarrow A$ satisfies weak monotonicity if it satisfies cyclical monotonicity for all cycles of length 2. That is, for all i, v_{-i} , and all v_i, v'_i , we have

$$v_i(g(v_i, v_{-i})) - v_i(g(v'_i, v_{-i})) + v'_i(g(v'_i, v_{-i})) - v'_i(g(v_i, v_{-i})) \geq 0.$$

5 Social Choice with Indifferences, Constrained Allocation

5.1 Introduction and the Equivalence of Constrained Allocation

Definition 5.1.1 (Social choice problem with indifferences). For each agent $i \in N$, suppose we have a partition $A^i := \{A_1^i, \dots, A_{m_i}^i\}$ of A such that if $x, y \in A_\ell^i$, then i is indifferent between x and y . Let P^i be the set of strict preferences on A^i . Then any $\succ_i \in P^i$ induces a weak ranking $\succ_i^*$ on A , defined by the following relation for $x \in A_\ell^i, y \in A_k^i$:

$$\begin{cases} x \succ_i^* y, & A_\ell^i \neq A_k^i \text{ and } A_\ell^i \succ_i A_k^i \\ x \sim_i^* y, & A_\ell^i = A_k^i \end{cases}.$$

A social choice function with indifferences is a function $f : \prod_{i \in N} P^i \rightarrow A$ and is defined by

$$\left(N, A, \left(A^i \right)_{i \in N} \right),$$

and the social choice problem is to select such a function given the indifference partitions.

Definition 5.1.2 (Constrained allocation problem). Suppose for each agent $i \in N$, we have a set X_i of available objects. An allocation is a function $\mu : N \rightarrow \cup_{i \in N} X_i$ such that for every $i \in N$, $\mu(i) \in X_i$. An allocation specifies for each agent i a feasible object to be received. Denote $\mathcal{A} := \{\mu : N \rightarrow \cup_{i \in N} X_i\}$ to be the set of allocations.

A constraint is a set $C \subseteq \mathcal{A}$ that is nonempty. Allocations in C are called *feasible*, while allocations not in C are *infeasible*. Let $\tilde{P}_i$ be the set of strict preferences on X_i . A mechanism is a function $f : \prod_{i \in N} \tilde{P}_i \rightarrow C$.

The constrained allocation problem is defined by

$$(N, (X_i)_{i \in N}, C).$$

Definition 5.1.3 (Strategy-proof, Pareto efficient). In the setting of a constrained allocation problem, a mechanism $f : \prod_{i \in N} \tilde{P}_i \rightarrow C$ is strategy-proof if there does not exist $i, \succ_i, \succ'_i, \succ_{-i}$ such that

$$f(\succ'_i, \succ_{-i})(i) \succ_i f(\succ_i, \succ_{-i})(i).$$

f is Pareto efficient if for all $\succ \in \prod_{i \in N} \tilde{P}_i$, there does not exist $\mu \in C$ such that:

- (i) For every $j \in N$, $\mu(j) \succ_j f(\succ)(j)$ and
- (ii) For some $i \in N$, $\mu(i) \succ_i f(\succ)(i)$.

5.2 Two Agent Constrained Allocation

Definition 5.2.1 (Local dictator). A local dictator at an infeasible allocation $(x, y) \notin C$ is an agent who gets to keep her top choice when x, y are top-ranked by agents 1 and 2 respectively.

6 Auctions

6.2 Overview of Auctions

Definition 6.2.1 (Bayesian game). A Bayesian game is defined by the following primitives:

- A set N of agents.
- For each $i \in N$, there is a set of types T_i which encodes her own type and her beliefs over others' types. We denote the type space $T := \prod_{i \in N} T_i$.
- For each $i \in N$, there is a set of actions A_i , and we denote the set of action profiles $A := \prod_{i \in N} A_i$.
- For each $i \in N$, there is a utility function
$$U_i : T_i \times A \rightarrow \mathbb{R}.$$
- (We assume there is) A common prior $\mu \in \Delta(T)$. Player i 's belief over the others' types is determined by conditioning μ on her realized type $t_i \in T_i$ and is notated $\mu(t_{-i} | t_i)$.

A strategy for player i is a function

$$\sigma_i : T_i \rightarrow \Delta(A_i).$$

The strategy profile $\sigma = (\sigma_i)_{i \in N}$ induces the payoff for agent i

$$U_i(\sigma, t_i) = \sum_{t_{-i}} \sum_{a \in A} u_i(t_i, a) \sigma_i(a | t_i) \sigma(a_{-i} | t_{-i}) \mu(t_{-i} | t_i).$$

The strategy profile σ is a Bayes-Nash equilibrium if for every agent i , σ , σ'_i , and t_i ,

$$U_i(\sigma, t_i) \geq U_i(\sigma'_i, \sigma_{-i}, t_i).$$

6.3 Revelation Principle

Definition 6.3.1 (Direct mechanism). A direct mechanism is a game where $\mathcal{A}_i = V$ so that agents report a value v .

6.4 Revenue-Maximizing Auctions

Definition 6.4.1 (Individual rationality). An auction game G with equilibrium s is individually rational if for each v ,

$$vX(v) - P(v) \geq 0.$$

Definition 6.4.2 (Regular value distribution, virtual value). A distribution with CDF F and density f is regular if the virtual value

$$c(w) := w - \frac{1 - F(w)}{f(w)}$$

is strictly increasing.

Definition 6.4.3 (Virtual value auction). In the virtual value auction, all buyers submit their values for the good. Using F , the seller converts these to virtual values. If no virtual value is positive, the seller keeps the good. Otherwise, she gives the good to the agent with the highest value. If there are multiple such agents, then each gets the good with equal probability. Buyers pay the lowest value they could have submitted and still won the auction with, i.e. the second-highest value after excluding duplicates.

8 Cooperative Games

8.2 Cooperative Games with Transferable Utility

Definition 8.2.1 (Cooperative game with transferable utility). A cooperative game with transferable utility is a pair (N, v) , where N is a set of agents and

$$v : 2^N \setminus \emptyset \rightarrow \mathbb{R},$$

where $v(S)$ is the surplus associated with every nonempty coalition S .

Definition 8.2.2 (Allocation). An allocation is a vector $x \in \mathbb{R}^n$ such that

$$\sum_{i \in N} x_i = v(N).$$

The allocation specifies the amount of surplus that each agents gets when the coalition includes all agents.

Definition 8.2.3 (Core). The core of (N, v) is the set of all allocations such that the sum of the allocation components within the coalition is larger than the surplus generated from the coalition:

$$\left\{ \text{allocations } x \in \mathbb{R}^n : \text{for all nonempty } S \subsetneq N, \sum_{i \in S} x_i \geq v(S) \right\}.$$

Alternatively and more clearly, it is

$$\left\{ x \in \mathbb{R}^n : \text{for all nonempty } S \subsetneq N, \sum_{i \in S} x_i \geq v(S) \text{ and } \sum_{i \in N} x_i = v(N) \right\}.$$

Definition 8.2.4 (Balanced weights). A set of balanced weights is a function $\delta : 2^N \setminus \{\emptyset, N\} \rightarrow [0, 1]$ such that for every agent $i \in N$,

$$\sum_{S: i \in S} \delta(S) = 1.$$

8.3 Cooperative Games with Non-Transferable Utility

Definition 8.3.1 (Core). An action $a \in A_N$ is in the core if there is no coalition $S \subseteq N$ and $a' \in A_S$ such that

$$a' \succsim_j a, \forall j \in S \text{ and } a' \succ_k a \text{ for some } k \in S.$$

9 Stable Matching

9.1 One-Sided Matching

Definition 9.1.1 (Matching). A matching is a function $\mu : N \rightarrow N$ such that $\mu \circ \mu = \text{id}_N$. Equivalently, if i is matched with j , then j is matched with i .

Definition 9.1.2 (Acceptable, individually rational). We say that j is acceptable to i if $j \succ_i i$. A matching μ is individually rational if $\mu(i)$ is acceptable for every i such that $\mu(i) \neq i$, i.e. if everyone who is matched weakly prefers his/her partner to him/herself.

Definition 9.1.3 (Stable, blocking pair). A matching μ is stable if it is individually rational and there is no pair i, j such that $i \neq j$ and

$$i \succ_j \mu(j) \text{ and } j \succ_i \mu(i).$$

A pair (i, j) in which

$$i \succ_j \mu(j) \text{ and } j \succ_i \mu(i)$$

is called a *blocking pair*, as they could form a coalition (blocking the matching μ) and both be better off.

9.2 One-to-One Two-Sided Matching

Definition 9.2.1 (Gale–Shapley (Deferred Acceptance Algorithm)). The traditional algorithm is the M -proposing deferred acceptance algorithm. The algorithm proceeds in the following steps:

- (1) Each $m \in M$ proposes to his favorite acceptable $w \in W$. Each $w \in W$ is tentatively matched with her favorite acceptable proposer and rejects everyone else. Call μ_1 this match. If there were no rejections, stop and return the tentative matches.
- (2) In each step $k \geq 2$, each $m \in M$ who was rejected in the previous round proposes to his favorite acceptable w who has not yet rejected him. Each $w \in W$ is tentatively matched with her favorite acceptable m among those who applied in this round and her tentative match from the previous round. All other applicants are rejected. Let μ_k denote this match. If there were no rejections, stop and return the tentative matches.

For any profile of preferences $\succ$, let $DA_M(\succ)$ be the matching returned by the M -proposing deferred acceptance algorithm.

Definition 9.2.2 (Achievable). We say that $w \in W$ is achievable for $m \in M$ if there exists a stable matching where m and w are matched.

Definition 9.2.3 (Mechanism). A one-to-one two-sided matching mechanism is a function $f : \mathcal{P} \rightarrow \Sigma$. It is stable if for every $\succ \in \mathcal{P}$, $f(\succ)$ is stable. In other words, mechanisms are stable if they give stable matchings for every preference profile.

9.3 Many-to-One Two-Sided Matching

Definition 9.3.1 (Matching). A matching is a function $\mu : M \rightarrow M \cup W$ such that:

- (i) $|\mu^{-1}(w)| \leq c_w$ for every w
- (ii) If $\mu(m) = m' \in M$, then $m = m'$.

Definition 9.3.2 (Stable, blocking pair). A matching μ is stable if:

- (i) Each m is matched to an acceptable w or is unmatched, and
- (ii) There is no $M' \subseteq M$ and some w such that $M' \succ_w \{m : \mu(m) = w\}$ and each $m \in M'$ weakly prefers w to his match in μ .

Such a pair (M', w) in (ii) is called a blocking pair.

Definition 9.3.3 (Responsive). A preference $\succ_w$ is responsive if there is a strict ranking $\succ_w^*$ on $M \cup \{w\}$ such that for any S with $|S| < c_w$,

- (i) For any $m, m' \notin S$, we have that

$$S \cup \{m\} \succ_w S \cup \{m'\} \iff m \succ_w^* m',$$

- (ii) and for every $m \notin S$,

$$S \cup \{m\} \succ_w S \iff m \succ_w^* w.$$

Definition 9.3.4 (Deferred acceptance algorithm, many-to-one). Let $\mu_0(m) = m$ for every m . The algorithm proceeds as follows:

- (i) All $m \in M$ apply to their favorite acceptable w . If for every m , no w is acceptable, return μ_0 . Otherwise, each $w \in W$ considers the set of applicants A_w^1 and rejects all $m \in A_m^1$ except those in

$$\arg \max_{\succ_w} \left\{ M' \subseteq A_w^1 : |M'| \leq c_w \right\}.$$

Let μ_1 be the matching in which m is matched to w if he applied to w and was not rejected, and where m is otherwise unmatched.

- (ii) For steps $k \geq 2$, each⁵ m applies to his favorite w who has not yet rejected him. If there are not applications, return μ_{k-1} . Each w gets applicants A_w^k and rejects all m not in

$$\arg \max_{\succ_w} \left\{ M' \subseteq A_w^k \cup \mu_{k-1}^{-1}(w) : |M'| \leq c_w \right\}.$$

Let μ_k be the matching where each m is matched to his favorite w to whom he applied and was not rejected. If there is no such w , then m is unmatched. If there are no rejections, then return μ_k .

9.4 More on the Deferred Acceptance Algorithm

Definition 9.4.1 (Aligned). We say that preferences $(\succ_m)_{m \in M}$ and $(\succ_w)_{w \in W}$ are aligned if there exists $U : M \times W \rightarrow \mathbb{R}$ such that for every $m \in M$ and $w \in W$, $U(m, \cdot)$ represents m 's preferences and $U(\cdot, w)$ represents w 's preferences.

10 Allocation Problems

10.1 Deterministic Allocation

Svensson (1999)

Definition 10.1.1 (Allocation, mechanism). An allocation is a bijection $\mu : N \rightarrow X$. We denote the set of allocations as A . A mechanism is a function $f : P^N \rightarrow A$.

Definition 10.1.2 ((Group) strategy-proof). A mechanism f is:

- Strategy-proof if for every i , $\succ \in P^N$, and $\succ'_i$,

$$f(\succ)(i) \succsim_i f(\succ'_i, \succ_{-i})(i).$$

- Group strategy-proof if for all $\succ \in P^N$, there is no nonempty coalition $S \subseteq N$ and joint misreport $\succ'_S = (\succ'_i)_{i \in S}$ such that

$$f(\succ'_S, \succ_{-S})(j) \succ_j f(\succ)(j) \quad \text{for all } j \in S$$

and

$$f(\succ'_S, \succ_{-S})(i) \succ_i f(\succ)(i) \quad \text{for some } i \in S.$$

- Monotonic if for any $\succ, \succ'$ such that for all i and all $x \in X$, if

$$f(\succ)(i) \succ_i x \implies f(\succ')(i) \succ'_i x,$$

then $f(\succ') = f(\succ)$. If the set of objects dispreferred to $f(\succ)(i)$ weakly grows from $\succ$ to $\succ'$ for agent i , then f is monotone if i maintains the same object.

- Non-bossy if for all i , $\succ_{-i}$, $\succ_i$, and $\succ'_i$,

$$f(\succ_i, \succ_{-i})(i) = f(\succ'_i, \succ_{-i})(i) \implies f(\succ_i, \succ_{-i}) = f(\succ'_i, \succ_{-i}).$$

For an agent i and preferences for the remaining agents, if i is able to change her preference without affecting her own allocation, then her changing her preference can't change others' allocations.

⁵Contrasts with the one-to-one DA- M algorithm.

Definition 10.1.3 (Serial dictatorship mechanism). The serial dictatorship mechanism with order $\rho : \{1, 2, \dots, |N|\} \rightarrow N$ (ρ is a bijection) is the mechanism that gives $\rho(1)$ her top choice, $\rho(2)$ her favorite choice of the remaining objects, and so on.

We call ρ the *picking order* of the mechanism.

Definition 10.1.4 (Neutrality). Let σ be a permutation on X . If $\mu \in A$, define $\sigma \mu$ to be the allocation where each agent i gets $\sigma(\mu(i))$, i.e.

$$(\sigma \mu)(i) := \sigma(\mu(i)).$$

If $\succ_i \in P$, let $\sigma \succ_i$ be the preference such that

$$x \sigma \succ_i y \iff \sigma^{-1}(x) \succ_i \sigma^{-1}(y).$$

A mechanism f is neutral if for every $\succ \in P^{|N|}$ and all permutations σ ,

$$f(\sigma \succ) = \sigma f(\succ).$$

10.2 Random Allocation

Definition 10.2.1 (Bistochastic matrix). A square matrix M is bistochastic if the rows and columns sum to 1 and all entries are positive, i.e.

$$\sum_j M_{ij} = 1, \forall i, \quad \sum_i M_{ij} = 1, \forall j.$$

Let $\mathcal{B}_{|N|}$ be the set of bistochastic $|N| \times |N|$ matrices.

Definition 10.2.2 (Random allocation mechanism). A random allocation mechanism is a function $f : P^{|N|} \rightarrow \mathcal{B}_{|N|}$. $f(\succ)_{ix}$ is agent i 's probability of being assigned house x when preferences are $\succ$ under f . We write $f(\succ)_i$ for the i th row of the matrix $f(\succ)$, which is agent i 's lottery over houses.

Definition 10.2.3 (First-order stochastic dominance). Let $p, p' \in \Delta(X)$ and let $\succ_i$ be a strict ordering on $X = \{x_1, x_2, \dots, x_{|N|}\}$. We say that p first-order stochastically dominates p' with respect to $\succ_i$ if

- $p(x_1) \geq p'(x_1)$,
- $p(x_1) + p(x_2) \geq p'(x_1) + p'(x_2)$,
- and so on, such that for every $x \in X$,

$$\sum_{y: y \succsim_i x} p(y) \geq \sum_{y: y \succsim_i x} p'(y).$$

We denote first order stochastic dominance of p over p' with respect to $\succ_i$ by

$$p \text{FOSD}_{\succ_i} p'.$$

Definition 10.2.4 (Strategy-proof, Ex-post efficient, Equal treatment of equals). Let $f : P^{|N|} \rightarrow \mathcal{B}_{|N|}$ be a random mechanism. f is strategy-proof if for all i , $\succ_{-i}$, $\succ_i$, and $\succ'_i$,

$$f(\succ_i, \succ_{-i})_i \text{FOSD}_{\succ_i} f(\succ'_i, \succ_{-i})_i.$$

Implicitly, if f is strategy-proof, then the lotteries $f(\succ_i, \succ_{-i})_i$ and $f(\succ'_i, \succ_{-i})_i$ must be comparable.

We say that f is ex-post efficient if for all $\succ$, $f(\succ)$ can be written as a convex combination⁶

$$\alpha_1 \mu_1 + \dots + \alpha_p \mu_p$$

where each μ_j is a deterministic Pareto efficient allocation.

f satisfies equal treatment of equals (ETOE) if for any i, j and $\succ_{-ij}$, if $\succ_i = \succ_j$,

$$f(\succ_i, \succ_j, \succ_{-ij})_i = f(\succ_i, \succ_j, \succ_{-ij})_j.$$

⁶We alternatively say that $f(\succ)$ has a social implementation $\sum_k \alpha_k \mu_k$.

Definition 10.2.5 (Random serial dictatorship, random top trading cycles). Random serial dictatorship (RSD) is the mechanism

$$\text{RSD}(\succ) = \frac{1}{|N|!} \sum_{\rho} \text{SD}_{\rho}(\succ).$$

Random top trading cycles (RTTC) is the mechanism

$$\text{RTTC}(\succ) = \frac{1}{|N|!} \sum_{\mu_0 \in A} \text{TTC}_{\mu_0}(\succ).$$

Note that $|N|! = |A|$, where A is the set of *allocations*, not objects.

Definition 10.2.6 (Ordinal efficiency). We say that f is ordinally efficient if for every $\succ$, there is no $M \neq f(\succ)$ such that

$$M_i \text{ FOSD}_{\succ_i} f(\succ)_i$$

for every i .

Definition 10.2.7 (Probabilistic serial mechanism). The probability serial mechanism is defined by the following:

- (i) Start at $t = 0$.
- (ii) As t increases, let agent “eat” probability shares of her favorite object at unit speed until one or more objects is fully consumed (i.e. the total probability share eaten of some object is 1). Continue by having every agent “eat” probability shares of her favorite object remaining until $t = 1$.

11 Information Design

11.2 General Model

Definition 11.2.1 (Signal structure). A signal structure is a collection of distributions $\{\pi(\cdot \mid \omega)\}_{\omega \in \Omega}$ where there is a finite set S such that each $\pi(\cdot \mid \omega)$ has support in S .

Definition 11.2.2 (Bayes-plausible). Given a prior $\mu_0, \tau \in \Delta(\Delta(\Omega))$ is Bayes-plausible if

$$\int \mu d\tau(\mu) = \mu_0.$$

A Tools

A.1 Systems of Linear Inequalities

Definition A.1.1 (Polyhedron, polytope). A polyhedron is the solution set to a system of linear inequalities $Ax \leq b$, and a polytope is a bounded polyhedron.

A.2 Fourier–Motzkin Elimination

Definition A.2.1 (Consequence relation). An inequality

$$a_0x \leq b_0$$

is a consequence relation of the system $Ax \leq b$ if every $x \in \mathbb{R}^n$ satisfying $Ax \leq b$ also satisfies $a_0x \leq b_0$.

A.6 Polyhedra

Definition A.6.1 (Face). A face of $P(A, b)$ is a set of the form

$$F = P(A, b) \cap \{x : a_0x = b_0\}$$

where $a_0x \leq b_0$ is a consequence relation of $Ax \leq b$.

Definition A.6.2 (Vertex). A face consisting of a single point is called a vertex.

Definition A.6.3 (Facet). A facet of $P(A, b)$ is a proper face $F \neq P(A, b)$ that is maximal among proper faces, i.e. there is no proper face $G \neq F$ with $F \subseteq G$.

A.7 Lattices and Tarski's Fixed Point Theorem

Definition A.7.1 (Partial order). A binary relation $\geq$ on a set L is a partial order if it is reflexive, transitive, and antisymmetric.

Definition A.7.2 (Complete lattice). A complete lattice $(L, \geq)$ is a set L with a partial order $\geq$ such that:

- (i) for every $A \subseteq L$, there is an element $u \in L$, denoted $\bigvee A$, such that $u \geq x$ for all $x \in A$ and whenever $v \geq x$ for all $x \in A$, we have $v \geq u$;
- (ii) for every $A \subseteq L$, there is an element $u \in L$, denoted $\bigwedge A$, such that $u \leq x$ for all $x \in A$ and whenever $v \leq x$ for all $x \in A$, we have $v \leq u$.

Definition A.7.3 (Order-preserving). If $(L, \geq)$ is a complete lattice and $f : L \rightarrow L$, we say f is order-preserving if

$$x \leq y \implies f(x) \leq f(y).$$